

Farhan Akbor Khan

Assignment - 03

Module-03

Software Testing Life Cycle (STLC) with AI

Q1. Create a Test Plan for an Online Shopping Website.

Answer:

Test Plan: Online Shopping Website

Aspect	Test Plan
What will be tested?	Registration/login, product search, product details, shopping cart, checkout, payment, order placement, order tracking, and logout.
How will testing be performed?	Functional, UI, usability, compatibility, security, and basic performance testing will be performed using manual and automated testing where appropriate.
Who will perform testing?	QA/Test Engineers will execute the tests. Developers will fix identified defects and the QA Lead will review the test results.
When will testing be performed?	Testing will be conducted throughout development, after feature completion, during integration and before release. Regression testing will be performed after bug fixes.
Which resources are required?	Test cases, test data, testing environment, browsers/devices, database, payment-test environment and bug-tracking/automation tools.
What risks and constraints exist?	Payment gateway failures, server downtime, security issues, limited test data/devices, third-party service failures and limited testing time.

Q2. Prepare a Test Checklist.

Create a checklist containing at least 20 validation points for the below features.

Login Checklist

Registration Checklist

Checkout Checklist

Answer:

Answer includes:

- Positive & negative testing
- Input validation
- UI/UX validation
- Security
- Cross-browser/device compatibility
- Session handling
- Payment failure scenarios
- Order and inventory validation

1. Login Checklist

Validation Point		Statu s
1	Verify login with valid email/username and password.	☐
2	Verify login fails with an incorrect password.	☐
3	Verify an appropriate error message is displayed for invalid credentials.	
4	Verify login fails with an unregistered email/username.	☐
5	Verify validation message appears when email/username is empty.	☐

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|----|---|--------------------------|
| 6 | Verify validation message appears when password is empty. | ☐ |
| 7 | Verify login cannot be submitted with invalid/empty input. | |
| 8 | Verify password is masked while typing. | ☐ |
| 9 | Verify the Show/Hide Password option works correctly. | ☐ |
| 10 | Verify the Remember Me option works as expected. | ☐ |
| 11 | Verify the Forgot Password link works correctly. | ☐ |
| 12 | Verify users are redirected to the appropriate landing page after successful login. | ☐ |
| 13 | Verify login works correctly across supported browsers. | ☐ |
| 14 | Verify login works correctly on mobile and desktop screens. | ☐ |
| 15 | Verify the user session is created after successful login. | ☐ |
| 16 | Verify logout terminates the user session. | ☐ |
| 17 | Verify protected pages cannot be accessed after logout using the Back button. | ☐ |
| 18 | Verify repeated failed login attempts are handled securely. | ☐ |
| 19 | Verify login credentials are transmitted securely over HTTPS. | ☐ |
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2. Registration Checklist

- | | Validation Point | Status |
|---|---|--------------------------|
| 1 | Verify registration with all valid required information. | ☐ |
| 2 | Verify registration fails when required fields are empty. | ☐ |
| 3 | Verify email format validation. | ☐ |
| 4 | Verify duplicate email registration is prevented. | ☐ |

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|----|---|--------------------------|
| 5 | Verify password meets the required complexity rules. | ☐ |
| 6 | Verify password and confirm-password fields match. | ☐ |
| 7 | Verify an appropriate error appears when passwords do not match. | ☐ |
| 8 | Verify password is masked while typing. | ☐ |
| 9 | Verify name fields accept valid characters and reject invalid input where applicable. | ☐ |
| 10 | Verify phone number format and length validation. | ☐ |
| 11 | Verify leading/trailing spaces are handled correctly. | ☐ |
| 12 | Verify maximum and minimum field lengths are enforced. | ☐ |
| 13 | Verify the Terms & Conditions checkbox is required if applicable. | ☐ |
| 14 | Verifying successful registration creates the user account correctly. | ☐ |
| 15 | Verify a confirmation/success message is displayed after registration. | ☐ |
| 16 | Verify email/OTP verification works if implemented. | ☐ |
| 17 | Verify the user can log in using the newly registered account. | ☐ |
| 18 | Verify registration works across supported browsers. | ☐ |
| 19 | Verify registration works correctly on mobile and desktop devices. | ☐ |
| 20 | Verify sensitive information such as passwords is handled securely. | ☐ |
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3. Checkout Checklist

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|---|--|--------------------------|
| 1 | Verify only selected products are displayed in the checkout. | ☐ |
| 2 | Verify product name, quantity and price are correct. | ☐ |

- 3 Verify product subtotal is calculated correctly. ☐
- 4 Verify shipping charges are calculated correctly. ☐
- 5 Verify applicable discounts/coupons are applied correctly. ☐
- 6 Verify invalid or expired coupons are rejected with an appropriate message. ☐
- 7 Verify the final order total correctly includes subtotal, shipping, discounts, taxes and other applicable charges. ☐
- 8 Verify required shipping address fields are validated. ☐
- 9 Verify valid phone numbers and postal codes are accepted. ☐
- 10 Verify users can select or add a shipping address. ☐
- 11 Verify all available shipping options, charges, and delivery times are displayed correctly. ☐
- 12 Verify that the selected payment method processes successful and failed transactions correctly. ☐
- 13 Verify invalid payment details are rejected appropriately. ☐
- 14 Verifying payment failure does not create a successful order. ☐
- 15 Verifying successful payment creates the order correctly. ☐
- 16 Verify a unique order ID is generated after successful order placement. ☐
- 17 Verify an order confirmation message/page is displayed. ☐
- 18 Verify order details appear correctly in the user's order history. ☐
- 19 Verify inventory/stock quantity is updated after successful purchase. ☐
- 20 Verify checkout is secure and sensitive payment information is protected. ☐